

Distribution-Substation and Primary-Feeder Planning

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Useful methods for distribution-substation and primary-feeder planning, based on the concept that actual distribution systems can be represented by a hypothetical system comparatively simple to construct and analyze, are presented. The methods can be used for quickly analyzing the service-quality and economic feasibility of planning alternatives or proposed changes.

ELECTRICAL-UTILITY distribution-system planning is extremely difficult because plans for the addition of new facilities must keep well ahead of the time when the exact locations and magnitudes of the future loads to be served by those facilities are known. This article presents a basis and methods for distribution-system planning which take load growth into account in such a way that exact locations and magnitudes of future loads need not be known to arrive at an economic system for serving the loads when they develop. To accomplish this, hypothetical equivalent distribution systems are devised for quickly analyzing the service quality and economic feasibility of distribution-system growth plans applicable to the load areas of the actual system being studied.

The system planning concepts and methods presented are particularly applicable to urban and suburban residential areas, served predominately by overhead circuits. In such load areas, some of the most fundamental planning problems center on the distribution substations, where the interrelated problems of subtransmission-system and primary-feeder design converge. Choice of distribution-substation arrangement, size, and voltage rating influences system planning in two directions: toward the source via the subtransmission circuits, and toward the load via the primary-feeder circuits.

The equivalent-system method recognizes the uncertainty of the exact magnitudes and locations of future loads in planning the locations and ratings of distribution substations, and the number and capability of the primary feeders required to distribute the power from these substations. In the equivalent system, each distribution substation with its primary-feeder circuits serves an area in which load density is comparable to the values existing or forecast for the actual system being studied. Results of studies made for

equivalent systems, for different values of load densities, are applicable to actual systems despite wide variations in the actual locations of individual loads in the equivalent or actual system.

HYPOTHETICAL EQUIVALENT SUBSTATION AREAS

THE HYPOTHETICAL equivalent distribution substation areas employed in these analyses have the shape of regular polygons. The number of sides to the polygon representing a load area is equal to the number of primary feeders radiating from the primary-circuit feed point, which is assumed to be located at the center of the area. The distribution substation may be located at the feed point, or may be remotely located, with one or more express primary feeders carried from the remote substation to the feed point. For example, a hypothetical area served by six circuits radiating out from a feed point would be hexagonally shaped, with each feeder carried from the center of the hexagon along an apothem or diagonal.

The use of hypothetical load areas and circuit patterns of symmetrical geometry greatly simplifies the comparison of alternative plans which may call for different sizes of distribution substation, different numbers of primary-feeder circuits, and different primary-feeder voltages. Such load areas and circuit patterns are abstractions because they do not exist exactly in such forms anywhere. These abstractions however, can be made to conform with the system they are devised to represent in all of the following ways: load density, substation size; substation or feedpoint service area in square miles; number of primary-feeder circuits, conductor material, and circuit impedance values; primary-feeder voltage.

Hypothetical distribution-substation areas cannot conform with actual systems in two ways: the exact duplication of circuit pattern, and the exact locations and magnitudes of loads. But as load grows, these two attributes of the system continually change, sometimes at so fast a rate that the system planner must spend all of his time with remedial measures or extensions in relatively small portions of the system, while the possibilities for system-wide engineering and economic benefits go uninvestigated.

DISTRIBUTION-SUBSTATION PLANNING

THE ONE PHENOMENON common to all distribution-system load areas is increasing load density. This is particularly true of the urban and suburban residential areas served by overhead primary-feeder circuits. Regardless of the plan adopted for serving such areas, load growth must be accompanied by, perhaps should be preceded by, an

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equivalent increase in distribution-substation and primary-feeder capacity. Very often, in order to achieve the required increases in system capability, changes in the basic system-growth plan must be made.

There are many features of the distribution substations and primary feeders that can be changed, over any reasonable period of time, provided that continuity of service to the customers is maintained while the system is being altered or expanded. Indeed many changes are very desirable, and serve to improve system economy, quality of service, and operation. Such changes may affect all of the following: primary-feeder voltage; distribution-substation size and arrangement; number of, and routing of primary-feeder circuits; primary-feeder loading; the various applications of supplementary voltage regulators, switched and unswitched capacitors, line-sectionalizing devices, etc. Any change must eventually lead to an increase in distribution-substation capacity.

The addition of distribution-substation capacity can follow virtually an infinite variety of general plans which must fall, however, between two extremes: In one extreme, the area served from a single substation-site remains fixed, and the capacity of the substation grows with load density. In the second extreme, a fixed amount of transformer capacity is located on a given substation site, and additional transformer capacity is located at new sites as the load grows. These two extremes of distribution-system growth will be called the constant substation-area plan and the constant-substation-size plan. Between these two extremes there is an infinite variety of plans calling for both the enlarging of existing substations and the addition of new substations.

As the load grows, the evolution of distribution substations and primary feeders may proceed slowly at certain times, and rapidly at other times, depending on what steps are taken. Changes which occur rapidly may be embodied into the system before the load has increased by very much.

During changes which evolve slowly, the load may increase appreciably.

Rate of load growth, with attendant system-wide increase of load density, will vary not only from one system to another, but also within individual substation areas. The engineering and economic factors determining the capability and pattern that a system should have at any given time depend on rate-of-load growth, which unfortunately is at times a highly uncertain factor. Consequently, in order to base a system plan on some limits within which the uncertainties of load growth will lie, the various steps that can be taken to increase distribution-substation and primary-feeder capacity can be divided into two general categories—those which are accomplished while load density remains constant and those usually achieved over considerably longer periods of time, during which load density increases. The accuracy of load growth forecast will determine the accuracy of the estimated time period over which such changes should be accomplished.

For short-range planning, the assumption can be made that load density remains constant. Even for short-range planning, however, any analysis which freezes load density will be recognized as not realistic, since load density in some urban and suburban residential areas increases appreciably even during time intervals measured in months. But a constant-density analysis, though it is academic, does serve the useful purpose of establishing certain limits on what can be accomplished when certain of changes are made.

SYSTEM CHANGES AT CONSTANT LOAD DENSITY

SYSTEM PLANNING often is centered about remedial measures which are required because of inadequate capability in existing distribution substations and primary feeders, or poor quality and continuity of service. Usually when distribution substations and primary-feeder circuits are inadequate from a load-carrying capability standpoint, voltage

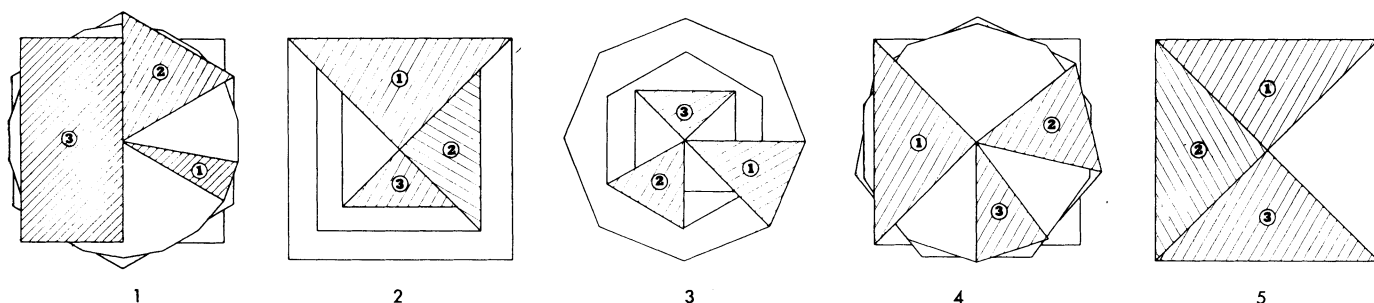


Fig. 1. Constant D , kva_{nt} , A_{nt} , V . The number of primary feeder varies as shown in equation 1. The relative values of primary-feeder voltage are: 1) $kv = 1.0$; 2) $kv = 1.73$; 3) $kv = 3.0$. **Fig. 2. Constant kva_{nt} , n , kv .** As load density increases while substation load remains constant, substation or feed point service area, and per-cent voltage drop in feeders vary as given by equation 6. Relative values of density, area and per-cent voltage drop are: 1) $D = 1.0$, $A_{st} = 1.0$, $V = 1.0$; 2) $D = 2.0$, $A_{st} = 0.5$, $V = 0.7$; 3) $D = 4.0$, $A_{st} = 0.25$, $V = 0.5$. **Fig. 3. Constant kva_{nt} , V , kv .** As load density increases while substation load remains constant, the number of primary feeders, and substation or feed-point area vary as given by equation 7. In this relative values of density, area and per-cent voltage drop are: 1) $D = 1.0$, $A_{st} = 1.0$, $V = 1.0$; 2) $D = 2.0$, $A_{st} = 0.5$, $V = 1.0$; 3) $D = 4.0$, $A_{st} = 0.5$, $V = 1.0$. **Fig. 4. Constant A_{nt} , V , kv .** As load density increases while substation or feed-point area, and per-cent voltage drop in feeders remain constant, the relationship between the variable system-planning parameters is given in equation 8. The relative values of density, number of primary feeders and per-cent voltage drop are: 1) $D = 1.0$, $n = 4$, $V = 1.0$; 2) $D = 2.0$, $n = 7$, $V = 1.0$; 3) $D = 4.0$, $n = 10$, $V = 1.0$. **Fig. 5. Constant A_{nt} , n , kv .** As load density increases while substation or feed-point area and number of feeder circuits remain constant, the relationship between the variable system-planning parameters is given in equation 9. In the above example, the relative values of density, number of primary feeders, and per-cent voltage drop are: 1) $D = 1.0$, $n = 4$, $V = 1.0$; 2) $D = 2.0$, $n = 4$, $V = 2.0$; 3) $D = 4.0$, $n = 4$, $V = 4.0$.

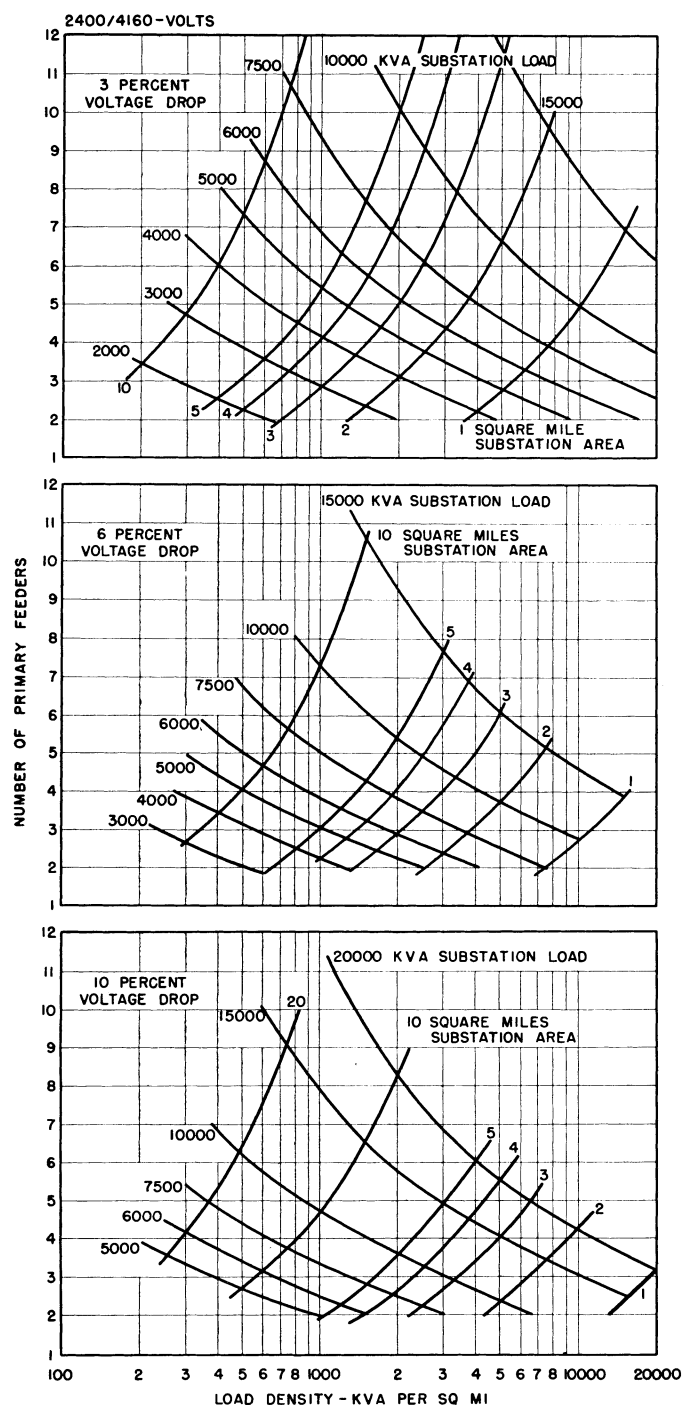


Fig. 6. The relationships between the system-planning parameters for systems employing 2,400/4,160-volt primary feeders. Number of primary feeders (n) vs load density (D) for 3, 6, and 10 per cent feeder voltage drop (V). Lines of constant substation or feedpoint loading (kva_{nt}) or constant substation or feedpoint service area (A_{nt}) are shown

drops are excessive, and outages are frequent and of long duration.

Some of the obstacles that may be encountered in attempting to achieve short-range remedies, with long-range continuing benefits, are: unavailability of new substation sites, and subtransmission and primary circuit easements; prohibiting ordinances or codes, or unfavorable civic sentiment toward more numerous or higher-voltage circuits.

Two basic types of changes to distribution substations and

primary feeders will be examined while load density remains constant: those involving an increase in primary-feeder voltage; and those for which primary-feeder voltage remains the same. Equations 1–5 describe five different plans for the changing of distribution-substation size and area, or the number of primary feeders, at constant load density. Equations 1, 2, and 3 deal with feeder-voltage increase. Equations 4 and 5 show what can be done through two remedial measures applied at the existing primary-feeder voltage. Hypothetical substation areas having the shape of regular polygons may be used to demonstrate the effects of these changes, as is demonstrated in Fig. 1.

Constant Load Density—Increasing Primary-Feeder Voltage.

Fig. 1 illustrates a type of system renovation that has achieved considerable popularity—increasing the primary-feeder voltage. When such an extensive and high-conversion-cost project is undertaken, there are certain factors which govern what can be accomplished. It is well known that for the same kilovolt-ampere loading, the per-cent voltage drop in a primary-feeder circuit will decrease inversely with the square of the voltage. A new voltage three times the original voltage will result in one ninth the original per-cent voltage drop. But when primary-feeder voltage is increased, optimum benefits may accrue if feeder loading or distribution-substation size is allowed to increase, at some sacrifice to the maximum possible reduction in feeder per-cent voltage drop. Fig. 1 describes a possible plan for increasing primary-feeder voltage.

At constant load density, constant substation size (and area), and constant per-cent voltage drop per feeder, the number of primary feeders varies inversely as the square of the primary-feeder voltage. The following relationship describes this effect:

$$n \propto \frac{1}{kv^2} \mid \text{Constant } D, kva_{nt}, A_{nt}, V \quad (1)$$

In Fig. 1, hypothetical distribution-substation areas employing three relative values of voltages are illustrated: 1.0 kv, 1.73 kv, and 3.0 kv, with 18, 6 and 2 primary feeders required at these voltages, respectively.

If for an increase in primary-feeder voltage, the number of primary feeders radiating from a substation or feed point remains the same, and the total per-cent voltage drop per feeder remains the same, then each feeder can serve a larger area at the same load density. For a constant number of feeders, and per-cent drop, at constant load density, the area served by each feeder is

$$kva_{nt} \propto A_{nt} \propto kv^{4/3} \mid \text{Constant } D, n, V \quad (2)$$

In equations 1, and 2, substation-revision plans provide for maximum benefits in reduction of number of feeders or increased substation size using the same number of feeders. However, optimum economic and service-quality benefits attendant with changeover to a higher primary-feeder voltage can usually be realized if use of the same, or somewhat fewer, primary feeders is accompanied by use of a larger distribution substation.

With changeover to a higher primary-feeder voltage, three results can be achieved: reduction in number of feeders; increased substation size; reduction in feeder per-cent voltage drop. The extent by which all these desirable

benefits can be accomplished simultaneously is given by the following relationship:

$$V \propto \frac{kva_{nt}^{3/2}}{n^{3/2}kv^2} \text{ Constant } D \quad (3)$$

Constant Load Density—Utilizing the Same Primary-Feeder Voltage. Changes in substation size and primary-feeder pattern while maintaining the same primary-feeder voltage do not usually require the numerous high-initial-cost system renovations which must accompany a changeover to a higher primary-feeder voltage.

At a constant load density, with the use of the existing primary voltage, an increase in the distribution-substation capacity at a given site can be accomplished only by increasing the number of primary feeders, or increasing the loading, and consequently per-cent voltage drop, in the existing feeders.

For example, an increase of 60 per cent in feeder loading results in twice the original per-cent voltage drop, and an increase of 110 per cent in feeder loading results in three times the original per-cent voltage drop. The following relationship describes this:

$$kva_{nt} \propto A_{nt} \propto V^{2/3} \text{ Constant } D, kv, n \quad (4)$$

An increase in per-cent voltage drop must ordinarily be compensated for by an increased use of supplementary forms of primary-feeder voltage regulation, so that the voltage spread allocated to the primary-feeder circuit does not become excessive. Thus if total drop is increased, additional voltage regulators must be installed at points along the primary feeder so that the drop between voltage-regulated points, or between the last point on the line at which a regulator is applied, and the end of the feeder, does not exceed the allowable spread voltage in the feeder.

A method for increasing the substation capacity at a given site, with the same primary-feeder voltage and per-cent voltage drop in feeders, is to increase the number of feeders. The expression for this type of change is approximately given by

$$n \propto \frac{kva_{nt}}{\tan^{1/3} \left(\frac{180}{n} \text{ degrees} \right)} \text{ Constant } D, kv, V \quad (5)$$

In equation 5 it will be noted that the tangent of half the angle represented by a feeder "sector" is included in the relationship. Thus the variation of kva_{nt} with n will depend to a considerable extent on the particular values of n employed before and after the change.

In a typical example of feeder addition at constant load density and per-cent drop, an increase from four to eight feeders permits a 50-per-cent increase in substation size. Tripling the number of feeders, from 4 to 12, permits the approximate doubling of substation size.

Summary: Constant Load-Density Changes. The relationships between substation size, and primary-feeder circuit pattern as changes are made at constant load density, have been shown. The extent to which these relationships are applicable will depend on the rapidity with which changes are accomplished relative to the continual system-wide increase in load density. During short periods of time, or in

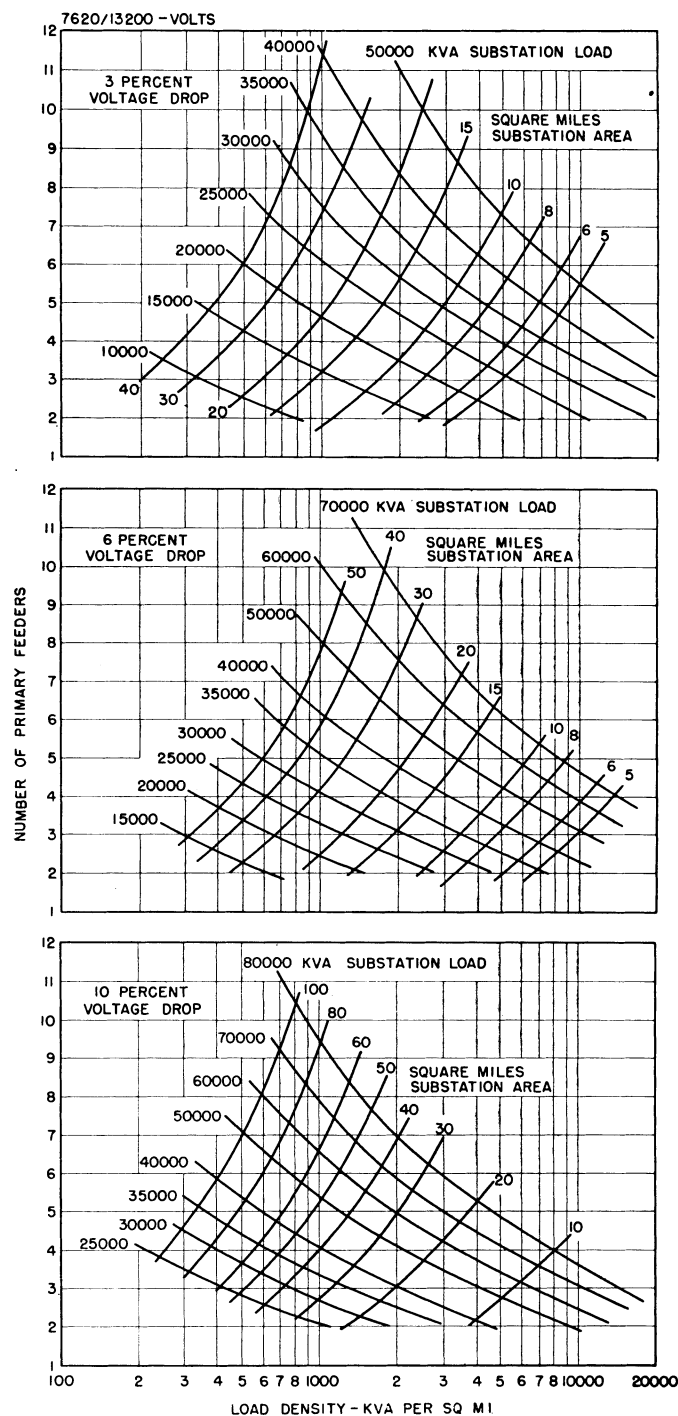


Fig. 7 Same as Fig. 6, except for systems employing 7,620/13,200-volt primary-feeder circuits

particular portions of the system in which load density increases only slightly, the relationships shown in equation 1-5 are applicable. But the basic long-range system planning problems deal with increasing load density.

SYSTEM CHANGES WITH INCREASING LOAD DENSITY

FOR MAKING SYSTEM CHANGES at increasing load density, the two extreme plans—one calling for constant substation size, and the other for constant substation area—have already been discussed. Neither type of plan permits the obtaining of maximum possible benefits in all components of the distribution system. A long-range optimum

plan for any specific system will generally be one in which the system will acquire some of the benefits of both plans, through continual compromises between the two extremes as load grows.

By the use of hypothetical substation areas and primary-feeder patterns, the changes that can be made in distribution-substation load areas as load density increases are shown in Figs. 2 through 5. Figs. 2 and 3 deal with the "constant substation-size plan," and Figs. 4 and 5 describe the "constant substation-area plan." In both plans, primary-feeder voltage remains the same as load density increases.

Increasing Load Density—The Constant Substation-Size Plan. If the amount of capacity located at any one substation site remains fixed as load density increases, then increments of new substation capacity that are continually added must be placed at new sites geographically located at points lying between existing substations. As load density increases, substation area decreases to keep substation loading constant. The principal disadvantage of such a plan is that as load grows, suitable substation sites may not be obtainable. However as one extreme representing a limit within which system growth must take place, this plan merits attention.

Constant substation-size plans may, in turn, be grouped into two basic schemes: In one, the number of primary-feeder circuits remains constant with increasing load density, but the per-cent voltage drop is high at the outset when load density is light and substation area is relatively large. In the second scheme, primary-feeder voltage drop remains constant with increasing load density, requiring the use of more feeders per substation at the outset, when density is low and feeder circuits are long. These two schemes are shown in Figs. 2 and 3.

With increasing load density, the use of a constant number of primary-feeder circuits supplied from a substation of constant size results in a continually decreasing per-cent voltage drop in feeder circuits as substation area decreases and primary feeders are shortened. The following relationship describes this behavior:

$$V \propto \frac{1}{D^{1/2}} \propto A_{nt}^{1/2} \quad \left| \text{Constant } kva_{nt}, n, kv \right. \quad (6)$$

In Fig. 2, when the load density in the area served by a constant-size substation has doubled and substation-area is halved, voltage drop is reduced to 70 per cent of its original value. When load density in the still further reduced substation area is four times the original value, per-cent voltage drop is reduced to half the original amount.

With increasing load density, and constant substation size, per-cent voltage drop in primary feeders can be kept constant while the required number of feeders per substation continually decreases. Thus construction of new feeder circuits can proceed at a slower rate than substation addition. The number of feeders per substation, as a function of load density, is given by the following relationships:

$$n \propto \frac{1}{D^{1/2}} \propto A_{nt}^{1/2} \quad \left| \text{Constant } kva_{nt}, V, kv \right. \quad (7)$$

In the illustration of Fig. 3, when the load density is doubled,

the number of primary feeders required to serve the smaller area containing the same load is reduced from eight to six. When load density is quadrupled, the number of feeders is reduced to four. Note that the need always to consider a whole number of primary feeders will require use of the nearest larger whole number of feeders if the value for n given by equation 7 does not result in a whole number.

Increasing Load Density—The Constant Substation-Area Plan. If the size of the area served from a given substation site remains constant as load density increases, then additional transformer capacity must be placed at that site as the load within the area increases. Additional primary-feeder capacity must also be continually provided to carry the power away from the growing substation. This represents another extreme in distribution-substation planning. The advantage of not having to locate substations at new sites as load grows is counterbalanced by the need for providing new primary-feeder circuits, or increasing the load-carrying capability of existing feeders.

As the load density in a fixed area served from one site increases, to keep the per-cent voltage drop in feeder circuits constant, additional feeders must be added. In the case shown in Fig. 4, when load density and substation load have doubled, the number of primary feeders increases from four to seven. At four times the original load density, ten primary are required. The relationship between number of feeders, and substation size, for the constant-area plan is given approximately by

$$n \propto \frac{kva_{nt}}{D^{1/3}} \quad \left| \text{Constant } A_{nt}, V, kv \right. \quad (8)$$

With substation area constant as load density increases, if more feeder circuits are not added, then the capability of the existing circuits must be more than adequate at the outset, or must be continually increased. A fixed area supplied by the same number of primary feeders as load density increases is shown in Fig. 5. The total per-cent voltage drop in each feeder increases however as the load grows, according to the following relationship:

$$V \propto kva_{nt} \propto D \quad \left| \text{Constant } A_{nt}, n, kv \right. \quad (9)$$

The capability of a constant number of feeders serving a constant area as load density increases, is limited in two ways: One limit is the thermal capability of the feeder conductors. The second limit is the maximum allowable per-cent voltage drop between voltage-regulated points on the feeder circuit. Provided that the existing feeder conductors are of adequate size to carry the load current, supplementary voltage regulators can be added along the feeders as load grows. Although the total voltage drop continually increases, this practice would keep the voltage spread along the feeders from becoming excessive.

Changes in Primary-Feeder Voltage With Load Density Increasing. Long-range system planning problems always involve increasing load densities. Often, such planning may consider increases in primary-feeder voltage. The increasing-density examples of equations 6 through 9 are based on unchanged primary-feeder voltage. The effects of increased primary-feeder voltage with increasing load density can be obtained by combining the plans of equations 1

through 5 with any one or more of those in equations 6 through 9. Such a procedure is one which "freezes" load density at several values, each larger than the preceding density. The effects of various plans can then be observed from one value of load density to the next. This technique of observing dynamic or growth situations through a number of successive static representations is a standard procedure in fields of endeavor as widely diverse as power system stability, economic theory, motion pictures, and theoretical physics.

PLANNING FOR SPECIFIC SYSTEMS OR AREAS

THE VARIOUS CHANGES by which distribution substations and primary feeders evolve are described in equations 1 through 9. The plans calling for such changes may be catalogued as follows:

1. Constant Load-Density Planning
 - (a) Increasing primary-feeder voltage: equations 1, 2, and 3
 - (b) Unchanged primary-feeder voltage: equations 4 and 5.
2. Increasing Load-Density Planning
 - (a) Constant-load distribution substations: equations 6 and 7.
 - (b) Constant-area distribution substations: equations 8 and 9.

The equations are gleaned from more comprehensive results of derivations. Since these equations deal with the physical and electrical attributes of the distribution system, substitution of wide ranges of practical values for the parameters into more complete equations yield information that can be used for the planning of specific systems. Typical results of such substitutions are shown in Figs. 6 and 7. All of the system-planning factors enumerated are represented quantitatively in Figs. 6 and 7. From these figures, the growth of a specific system following any one of the plans can be observed.

Figs. 6 and 7 are, essentially, plots showing the number of primary feeders (n) per substation or feed point versus load density (D) within the substation or feed-point service area. The variation of n with D is shown for constant substation or feed-point loading (kva_{n1}), or constant substation or feed-point area (A_{n1}). Fig. 6 deals with a primary-feeder voltage of 2,400/4,160 volts. Fig. 7 shows the relationships between the planning factors at 7,620/13,200 volts primary-feeder voltage. Each set (frame) of curves represents a different total per-cent voltage drop (V) in primary feeders. Curves for 3-, 6-, and 10-per-cent voltage drop are shown.

The system-planning factors, namely D , kva_{n1} , A_{n1} , n , kV , and V , may be followed by using these curves while one or more of the factors is kept constant, or allowed to vary. The interrelated changes in magnitude or quantity of the six parameters are given in Figs. 6 and 7. An important system-planning ingredient not included in these relationships is "time." During intervals of time over which load density increases little in certain portions of the system, the constant-density approach may be adopted for making changes. In other portions of the system, or over other

periods of time during which load and density grow appreciably, the increasing-density plans may be applied.

SUMMARY AND CONCLUSIONS

USEFUL METHODS for distribution-substation and primary-feeder planning have been given. The planning methods are based on the concept that actual distribution systems can be represented by hypothetical systems which are relatively simple to construct and analyze. For the purposes of system planning, these hypothetical systems are equivalent to the actual system which they represent.

Analysis of a number of primary-feeder circuit patterns of different geometric arrangement has shown, in related studies, that distribution-substation and primary-feeder planning based on such circuit patterns are applicable to actual distribution systems. Although the ultimate locations and magnitudes of future loads will shape the growth of every distribution system, they cannot be ascertained far enough in advance to permit the exact determination of the distribution-system facilities required to serve these loads.

The exact locations of individual loads in urban-suburban areas need not be considered in studying the interrelated effect of the system-planning parameters which are: load density; distribution-substation size; primary-feeder voltage; number of feeders per substation; allowable voltage drop in feeder circuits.

The study of equivalent systems brings out the reasons for, and nature of, the compromises that must be made in distribution-system design. Several plans for changes in distribution-substation size, number of primary-feeder circuits, and feeder loading, as load density remains constant or increases, are shown (Figs. 1 through 5). The construction of these hypothetical substation areas simplifies evaluating the effects of changes made in actual distribution systems. The quantitative relationships between the system-planning parameters are derived, and are illustrated in generalized curves, (see Figs. 6 and 7). These relationships can be used for quickly analyzing the service-quality and economic feasibility of planning alternatives or proposed system changes.

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Nomenclature

- D = Load density in kilovolt-amperes per square mile
 n = Number of primary-feeder circuits emanating from a distribution substation site, or primary feed point
 kva_n = The kilovolt-ampere loading per feeder
 kva_{n1} = Total kilovolt-ampere loading for all feeders of an n -feeder substation
 A_n = Area in square miles served by one primary feeder
 A_{n1} = Total area in square miles served by an n -feeder substation
 kV = The voltage at which a primary feeder operates
 V = The total percent-voltage drop in a primary-feeder circuit